

MERIDA INTL, Mexico

WMO: 766440

Lat: 20.9469N

Lon: 89.6522W

Elev: 11

StdP: 101.19

Time Zone: -6.00 (MEX)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.8	15.7	9.8	7.5	21.9	11.0	8.2	22.1	10.2	26.5	9.4	26.7	1.0	110	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.0	38.8	24.4	37.4	24.5	36.4	24.6	28.9	32.6	28.1	31.8	27.5	31.3	4.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.1	24.3	30.6	27.2	23.0	29.5	26.8	22.4	29.1	94.4	32.4	90.2	30.7	87.1	31.6	34.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
9.7	8.4	7.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			8.9	40.6	1.5	1.3	7.8	41.6	7.0	42.3	6.1	43.0	5.1	44.0
			8.4	30.4	1.7	1.5	7.2	31.4	6.3	32.3	5.3	33.1	4.1	34.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.4	23.8	25.4	26.9	29.0	30.2	29.5	29.1	29.1	28.8	27.4	25.3	24.4
	DBStd	2.97	2.76	2.82	2.86	2.38	1.79	1.51	1.09	1.03	1.25	1.83	2.24	2.59
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	1	1	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6358	429	431	524	569	626	584	592	593	563	538	460	447
	CDD18.3	3317	172	198	266	319	368	334	334	334	313	280	210	189
	CDH23.3	33557	1298	1763	2716	3673	4440	3670	3512	3511	3183	2640	1727	1424
	CDH26.7	15973	505	814	1396	2018	2450	1808	1688	1668	1446	1049	611	519
Wind	WSAvg	3.4	3.3	3.8	4.3	4.5	4.3	3.7	3.2	2.8	2.6	2.8	2.8	2.9
Precipitation	PrecAvg	1018	32	28	25	33	61	160	147	154	171	117	49	40
	PrecMax	1274	132	151	63	127	127	306	234	371	304	298	150	129
	PrecMin	669	0	3	3	0	17	38	52	63	54	26	10	5
	PrecStd	148	35	33	16	33	36	62	44	61	67	60	35	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	36.2	38.2	40.2	40.2	38.4	37.2	37.1	37.2	35.9	34.1	34.0
		MCWB	23.0	22.7	23.5	24.2	24.4	25.5	25.6	26.6	26.9	26.2	25.0	24.2
	2%	DB	32.9	34.7	36.8	38.2	38.9	37.0	35.9	35.8	35.2	34.0	32.8	32.2
		MCWB	22.8	22.9	23.3	23.9	24.3	25.6	26.1	26.4	26.6	26.2	24.6	23.6
	5%	DB	31.4	33.2	35.2	37.0	37.6	35.8	34.8	34.7	34.1	32.9	31.3	31.1
		MCWB	22.6	22.5	22.9	23.5	24.4	25.7	26.1	26.3	26.6	26.1	24.3	23.4
	10%	DB	29.8	31.9	33.8	35.8	36.1	34.1	33.8	33.7	33.0	31.9	30.1	29.9
		MCWB	22.3	22.4	22.6	23.4	24.6	25.7	25.9	26.2	26.3	25.6	23.7	23.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.1	26.3	26.8	27.8	28.7	29.2	29.4	29.7	29.8	29.0	27.7	27.2
		MCDB	29.3	31.6	32.4	33.4	34.3	33.0	33.2	33.6	33.4	31.9	30.3	30.2
	2%	WB	24.7	24.8	25.2	26.2	27.1	27.9	28.1	28.4	28.5	27.7	26.3	25.5
		MCDB	29.1	29.9	31.8	32.5	32.4	31.7	31.7	32.1	32.0	30.9	29.7	29.2
	5%	WB	23.9	24.1	24.5	25.5	26.3	27.1	27.2	27.6	27.6	27.0	25.7	24.7
		MCDB	28.5	29.1	30.8	32.0	32.2	31.1	31.1	31.5	31.2	30.5	29.2	28.7
	10%	WB	23.2	23.4	23.8	24.8	25.7	26.6	26.7	26.9	27.0	26.3	24.9	24.0
		MCDB	27.2	28.2	29.8	31.5	31.9	30.9	30.9	31.1	30.8	29.9	28.5	27.6
Mean Daily Temperature Range	5% DB	MDBR	11.7	12.7	13.5	13.8	13.0	11.0	11.1	11.0	10.5	9.8	10.6	11.1
		MCDBR	13.4	14.3	15.2	15.3	14.6	12.6	12.0	11.9	11.6	10.7	11.3	12.1
		MCWBR	5.6	5.1	4.9	4.5	4.3	4.5	5.0	5.1	5.1	4.7	5.2	5.5
	5% WB	MCDBR	11.6	12.5	13.7	13.6	13.0	11.1	11.2	11.2	10.7	9.8	9.8	10.6
		MCWBR	5.8	5.3	5.2	5.1	4.9	4.9	5.4	5.5	5.4	5.0	5.3	5.6
Clear-Sky Solar Irradiance	taub	0.358	0.372	0.400	0.435	0.451	0.448	0.465	0.441	0.411	0.395	0.375	0.371	
	taud	2.554	2.509	2.402	2.292	2.292	2.348	2.283	2.366	2.439	2.486	2.505	2.521	
	Ebn at Noon	918	923	906	873	850	845	832	857	883	887	892	890	
	Edh at Noon	94	104	120	136	135	126	135	125	114	105	98	95	
All-Sky Solar Radiation	RadAvg	4.12	4.92	5.79	6.35	6.41	5.94	6.47	6.31	5.71	5.08	4.46	4.00	
	RadStd	0.29	0.37	0.27	0.30	0.32	0.36	0.27	0.23	0.32	0.24	0.21	0.26	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	+0.88	+1.06	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	+0.88	+1.06	N/A	N/A	N/A	N/A

Nomenclature: See separate page